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A0 · Reverse Engineering Three Top Exemplar Papers and a Five-Round Thinking Drill Report

三篇范本逆向拆解与五轮思维演练

TwinsEarth · 智能体宇宙论文系列



A0 • Reverse Engineering Three Top Exemplar Papers and a Five-Round Thinking Drill Report

Version: UDOS Full Paper Text v7.5.0 / Stage A (exemplar reverse engineering) Date: 2026-09-20 (Asia/Shanghai, UTC+8) Methodology statement: this report is **not a recitation from memory**. All three PDFs were extracted page by page with PyMuPDF (PyMuPDF 1.28.2 / MuPDF 1.28.2) using `get_text("text", sort=True)`, and key figure/table pages were rendered with `get_pixmap` and read image by image for cross-checking (axes, three-line tables, layout). All numbers, section numbers, figure numbers come from the PDF original text, no extrapolation. Extraction artifacts retained: `_work/kaplan.txt` (30 pages / 124,762 chars), `_work/alexnet.txt` (9 pages / 37,959 chars), `_work/nsa.txt` (24 pages / 76,441 chars), and rendered images `_work/{alex,kap,nsa}_p*.png`.

0. Quick Overview of the Three Exemplars (one-sentence positioning)

Short name	True bibliographic record (PDF measured)	Length/layout	Role in this report
AlexNet	Krizhevsky, Sutskever, Hinton. ImageNet Classification with Deep Convolutional Neural Networks. NeurIPS 2012. 9 pages	two-column conference layout	The " engineering detail is the scientific contribution " paradigm: every trick paired with a quantified ablation + error-rate comparison table
Kaplan	Kaplan, McCandlish, Henighan et al. Scaling Laws for Neural Language Models. arXiv: 2001.08361v1, 2020. 30 pages	single-column arXiv + explicit contents + Appendices A–D	The " controlled variables + power-law fitting + trend extrapolation " paradigm: conclusions first, exponent table at the end, a dedicated Caveats section
NSA	Yuan, Gao, Dai ... Ruan, Zhang, Liang, Zeng. Native Sparse Attention: Hardware-Aligned and Natively Trainable Sparse Attention. arXiv:2502.11089v1, 2025 (DeepSeek-AI × Peking University × U.W.). 24 pages	single-column LaTeX conference layout + three-line tables	The " hardware constraint → architecture design → end-to-end trainable → layer-by-layer/task-by-task comparison " paradigm

Important erratum (double-confirmed by user instruction and the PDF original): the attachment DeepSeek论文2502.11089v1.pdf is measured to be titled **Native Sparse Attention (NSA)**, **not DeepSeek-V3, nor DeepSeek-R1**. R1/V2/R1 appear only in its references and comparison baselines. This report and the master library uniformly use NSA.

Part One • Paper-by-Paper Reverse Engineering

I. AlexNet (Krizhevsky/Sutskever/Hinton, NeurIPS 2012)

1.1 Structural Skeleton and Length Allocation

The whole paper is 9 pages, no explicit contents, two columns. Sections and lengths measured as follows:

Section	Title	Pages	Length feature
—	Abstract	p1	single paragraph, task + headline number + technique + results all packed into one paragraph
1	Introduction	p1–p2	from "the small-dataset era" to "why large models + CNN," ending with an explicit list of 5 specific contributions
2	The Dataset	p2	data split, preprocessing (256 × 256 downsampling, mean subtraction)
3	The Architecture	p3–p5	3.1 ReLU / 3.2 Multi-GPU / 3.3 LRN / 3.4 Overlapping Pooling / 3.5 Overall. The original text explicitly says "3.1–3.4 are ordered by our estimated importance, most important first"
4	Reducing Overfitting	p5–p6	4.1 Data Augmentation / 4.2 Dropout
5	Details of learning	p6	SGD hyperparameters, initialization, learning-rate schedule
6	Results	p7	Table 1/2 + Fall2009; 6.1 Qualitative Evaluations (Fig3/Fig4)
7	Discussion	p8	convergence
—	References	p9	[1]–[26], numbered

Allocation observation: methods (3+4+5) about 45%, results (6) about 15%, introduction (1) about 20%. **Introduction and methods are nearly equal in length**—typical of an "engineering-contribution" paper: the contribution itself is a string of engineering decisions, the introduction needed to explain "why each decision matters."

1.2 Abstract Writing (sentence by sentence)

The abstract is one paragraph, about 200 words, fixed seven beats: 1. **Task:** "We trained a large, deep CNN to classify the 1.2 million ... LSVRC-2010 ... into 1000 classes." 2. **Headline number:** "top-1 and top-5 error rates of **37.5% and 17.0%** ... considerably better than the previous SOTA." 3. **Scale anchor:** "60 million parameters and 650,000 neurons ... 5 conv + 3 FC + 1000-way softmax." 4. **Speedup means:** "non-saturating neurons and a very efficient GPU implementation." 5. **Anti-overfitting means:** dropout. 6. **Competition result (second hammer):** "ILSVRC-2012 winning top-5 **15.3%**, compared to **26.2%** second-best."

Pattern: **task** → **dual-indicator headline numbers** → **scale parameters** → **two technical keywords** → **one competition verdict**. All numbers go into the abstract.

1.3 Introduction Argumentation Pattern

- **Trend argument:** small datasets (NORB/Caltech/CIFAR) → real objects varied → must have larger data → ImageNet 15M/22k.
- **Motivation argument:** tasks too complex to be fully specified by data → need priors → CNN's locality/stationarity priors fit exactly.
- **Feasibility argument:** CNN previously "prohibitively expensive" → modern GPUs + optimized convolution make it feasible.
- **Contribution list:** ending "The specific contributions of this paper are as follows:" introducing 5 items (largest model, open-source GPU implementation, new structural features, anti-overfitting, depth importance).

- **Honest closing:** "network's size is limited mainly by GPU memory and training time ... **All of our experiments suggest results can be improved simply by waiting for faster GPUs and bigger datasets.**"

1.4 Related Work Organization

AlexNet **has no independent Related Work section**—related work is scattered into the introduction and the "we are not the first" paragraphs of each technical subsection (e.g. 3.1 cites Jarrett et al. and distinguishes that they observed anti-overfitting while we observe speedup). Common practice for 2012 conference papers, contrasting with Kaplan/NSA's independent Related Work.

1.5 Division of Methods/Experiments/Results/Discussion

- **Methods = contribution list:** 3.1–3.4 each trick one section, **each section ending with the error reduction that trick brings** (see 1.6).
- **Experiments:** Section 5 only writes "how trained" (hyperparameters), no results; results concentrated in Section 6.
- **Results:** using comparison tables (Table 1/2) + one cross-dataset (Fall2009) + qualitative evaluation.
- **Discussion:** two paragraphs, stressing "depth really matters" (removing any conv layer -2%) and "did not use unsupervised pretraining but expect it useful."

1.6 Transferable Argumentation Paradigms (this paper distills 9 items)

1. **Engineering detail is the scientific contribution:** ReLU, dual-GPU split, LRN, overlapping pooling, dropout, PCA color perturbation—all packaged as novel features, each with a name.
2. **Every trick paired with a quantified ablation** (verbatim from the original): - Dual-GPU split: "reduces top-1/top-5 by **1.7% and 1.2%**"; - LRN: "reduces by **1.4% and 1.2%**"; on CIFAR-10 13% $\rightarrow$ 11%; - Overlapping pooling ($s=2, z=3$ vs $s=z=2$): "reduces by **0.4% and 0.3%**"; - PCA color augmentation: "reduces top-1 by **over 1%**"; - dropout: "roughly doubles the number of iterations to converge" (cost also written).
3. **Fairness self-disclosure:** footnote 2 actively admits the one-GPU comparison "is biased in favor of the one-GPU net, since it is bigger"—**actively disclosing a bias against oneself**, in exchange for reviewer trust.
4. **Using ablations to prove "depth is indispensable":** "removing any convolutional layer (each $\leq 1\%$ of params) resulted in inferior performance," quantified again in the discussion as "middle-layer removal $\approx -2\%$ top-1."
5. **Dual-column error-rate comparison tables:** Table 1/2 give both top-1 and top-5; others' results in italics, own in roman; cross-dataset sub-tables (2010 / 2012 / Fall2009).
6. **Qualitative figures supplement quantitative gaps:** Fig3 shows learned kernels (GPU1 color-invariant vs GPU2 color-dependent, **and independent of initialization**); Fig4 uses 4096-dim features for nearest-neighbor retrieval, proving "high-level features are semantic, not pixel."
7. **Training details complete to reproducibility:** batch 128, momentum 0.9, weight decay 0.0005, weight init Gaussian std=0.01, ReLU bias=1, LR 0.01 manually divided by 10 three times, ~ 90 epochs, $2 \times \text{GTX580}$ for 5–6 days.
8. **Counterintuitive findings written as selling points:** "weight decay here is **not merely a regularizer**: it reduces the training error."

9. **Test-time ensembling at zero extra training cost:** 10-crop (four corners + center + horizontal flip) averaging, training-time CPU augmentation/GPU training in parallel = "computationally free."

1.7 Figure/Table List (one by one, with axes and self-consistency)

Figure/ Table	What it conveys	Axes/baseline/confidence	Caption self-consistency
Fig 1	ReLU vs tanh convergence speed	line chart, vertical training error rate, horizontal Epochs; solid ReLU/dashed tanh; caption explicitly says "learning rates independently tuned to fastest, no regularization"	high: caption carries the experiment setup and applicability boundary ("effect varies with architecture")
Fig 2	Overall architecture	3D isometric schematic, labeling per-layer neuron counts 253440–186624–64896–64896–43264–4096–4096–1000; two-GPU division of labor	high: caption explains at which layers the two GPUs communicate
Fig 3	Learned filters	96 $11 \times 11 \times 3$ kernel visualizations; upper 48 GPU1 / lower 48 GPU2	high: directly corresponds to the § 6.1 specialization observation
Fig 4	Qualitative evaluation	left: 8 images + top-5 predictions; right: 5 query images + 6 nearest-neighbor training images	high: explains the similarity metric (4096-dim Euclidean distance)
Table 1	ILSVRC-2010 test set comparison	three rows: Sparse coding 47.1/28.2, SIFT+FVs 45.7/25.7, CNN 37.5/17.0 ; others' best in italics	high: caption explains the italics meaning
Table 2	ILSVRC-2012 validation+test	columns: Top-1(val)/Top-5(val)/Top-5(test); 1CNN/5CNNs/1CNN/7CNNs	high: the * pretraining has a footnote explanation

Note: AlexNet's tables are plain two-column small tables, **not modern booktabs three-line tables**. It conveys comparison through "italics = others, roman = self, numeric columns aligned." This forms an era contrast with NSA's three-line tables.

1.8 How to Write Data Validation Methods

- **Split:** ILSVRC = ~1.2 million training / 50k validation / 150k test; explicitly says "ILSVRC-2010 is the only version with test labels, hence main experiments done on 2010."
- **Metric definition:** top-5 error defined on the spot (the share where the correct label is not in the top five predictions).
- **Compute budget:** 2 × GTX580 3GB, 5–6 days, 90 epochs.
- **Reproduction config:** see 1.6 item 7, all written.
- **Comparison:** using contemporaneous SOTA (sparse coding ensembles, Fisher Vectors) as baselines, not self-comparison.

1.9 Layout Standards

Two-column NeurIPS proceedings; sections Arabic numerals (1–7); references numbered [n], by order of appearance; formulas inline (e.g. LRN formula, SGD update); footnotes use on-page numbers (footnote 1 gives the code link, footnote 2 discloses bias, footnote 3 points to parameter files).

II. Kaplan 2020 Scaling Laws for Neural Language Models

2.1 Structural Skeleton and Length Allocation

The whole paper is 30 pages, **single-column arXiv**, **Contents explicitly printed before the body**. Sections measured:

Section	Title	Pages	Feature
1	Introduction (1.1 Summary=8 conclusion bullets; 1.2 Summary of Scaling Laws=formulas; 1.3 Notation)	p1–6	all conclusions first, body then proves
2	Background and Methods (2.1 parameter count/ compute; 2.2 training process; 2.3 datasets)	p6–7	
3	Empirical Results and Basic Power Laws	p7–10	shape-invariant, N scaling, LSTM comparison, cross-distribution generalization, D/C
4	Infinite Data Limit & Overfitting	p10–12	$L(N,D)$ equation + three construction principles
5	Scaling with Model Size & Training Time	p12–14	Bcrit correction, $L(N, S_{min})$, early-stopping lower bound
6	Optimal Allocation of Compute Budget	p14–17	theoretical derivation vs empirical comparison; 6.3 self-found contradiction
7	Related Work	p18	independent section
8	Discussion	p18–19	ideal gas law analogy closing
A–D	Appendices: A power-law summary / B compute frontier derivation / C Caveats (6 items) / D supplementary figures	p20–30	a dedicated "limitations" appendix

Allocation observation: methods and empirics (2–5) about 60% of the body, but **the introduction (including 1.1/1.2) itself already states all quantitative conclusions**. Appendices (A–D) take 10 pages, about 1/3, carrying all summary tables and supplementary experiments—the main text "clean," appendices "solid."

2.2 Abstract Writing

Single paragraph about 150 words, core sentence: "The loss scales as a **power-law** with model size, dataset size, and compute ... with some trends spanning **more than seven orders of magnitude** ... architectural details ... minimal effects ... **Larger models are significantly more sample-efficient** ... optimally compute-efficient training involves training very large models on a relatively modest amount of data and stopping significantly before convergence."

Pattern: **one sentence on the paradigm (power-law) + one sentence on the range (>7 orders of magnitude) + one counterintuitive conclusion (large models more efficient, no need to train to convergence).**

2.3 Introduction Argumentation Pattern (1.1 is the exemplar of exemplars)

1.1 Summary uses **8 bullets** to first throw out all headline conclusions, **each ending with a parenthetical source section** (e.g. "(Section 3)"), forming a "conclusion map": - Performance depends strongly on scale, weakly on model shape; - Smooth power laws (>6 orders of magnitude, no sign of upper-end deviation); - Universality of overfitting ($N^{0.74}/D$ determines the penalty); - Universality of training (early-curve extrapolation predicts long training); - Transfer improves with test performance (constant offset); - Sample efficiency (large models save samples); - Convergence is inefficient (should early-stop); - Optimal batch size ($\approx 1-2M$ tokens).

1.2 immediately gives **all closed-form equations** (1.1)–(1.8), 1.3 centrally defines symbols ($L, N, C, D, B_{crit}, C_{min}, S_{min}, \alpha_X$).

2.4 Related Work Organization

Section 7 independent, clustered by "intellectual origin": multiple sources of power laws, early dataset-size power laws, **explicit disagreement** with [HNA+17] (they are superlinear, this paper sublinear $D \propto N^{0.74}$), EfficientNet comparison, over-parameterized generalization (jamming transition, not observed). **Dares to name disagreements.**

2.5 Division of Methods/Experiments/Results/Discussion/Limitations

- Methods = engineering bookkeeping: Table 1 computes per-layer parameters and FLOPs; explicitly "excluding embeddings yields the clean trend."
- Results = figures + fits: every figure is "scatter + power-law fit line."
- Discussion = analogies and outlook: ideal gas law / "more is different."
- **Limitations = dedicated Section C:** 6 Caveats all bulletized.

2.6 Transferable Argumentation Paradigms (this paper distills 10 items)

1. **Controlled variables + power-law fitting + trend extrapolation:** fix N vary shape (Fig5), fix shape vary N ; $L(N) = (N_c/N)^{\alpha}$.
2. **Taking "range" as the core selling point:** >7 orders, >6 orders repeatedly appear, using span to strengthen credibility.
3. **Conclusions first (conclusion map):** 1.1 eight bullets + parenthetical sections, the body is the "proof map."
4. **Parameter-table-driven reproducibility:** Table 1 per-layer parameters/FLOPs; Appendix A Tables 4/5/6 **centralize all exponents and scaling constants** ($\alpha_N=0.076$, $\alpha_D=0.095$, $\alpha_C=0.057$, $\alpha_{min}=0.050$, $\alpha_B=0.21$, $\alpha_S=0.76$; $N_c=8.8e13$ etc.) **into a lookup table.**
5. **Theory→empirics bidirectional reconciliation:** from $L(N, S_{min})$ analytically deriving $\alpha_{minC} \approx 0.054$ (Eq 6.4), then comparing with measured ≈ 0.050 ("excellent agreement to within a few percent").
6. **Counterintuitive conclusions amplified:** "Big models may be more important than big data."; compute-efficient training only needs to stop at " $\sim 10\%$ above convergence loss" ($\alpha_N/\alpha_S \approx 0.1$); data sublinear growth $D \propto N^{0.74}$.
7. **Actively finding one's own contradiction:** 6.3 points out $L(C_{min})$ and $L(D)$ must intersect at $C \sim 10^4$ PF-days, $N \sim 10^{12}$, $L^* \sim 1.7$ nats = the law will fail, explicitly saying "numbers highly uncertain, magnitude floating up to an order of magnitude."

8. **Dedicated Caveats:** ① no theory; ② no confidence in Bcrit extrapolation; ③ poor fit in the small-data region (epoch only 40 steps); ④ compute estimates ignore the nctx term; ⑤ may have missed tuning init/momentum; ⑥ learning rate sensitive to target loss.
9. **Symbol and definition discipline:** 1.3 centrally defines; stresses N as "non-embedding parameters," and uses Fig6 left-right comparison to prove "with embeddings the trend is polluted"—**turning the measurement definition itself into a methodology figure.**
10. **Honest fit-quality self-assessment:** "the fits are imperfect ... quite compelling given the simplicity of Equation (5.6)"; explicitly the 1-layer network excluded from the fit (the "lump" in Fig13).

2.7 Figure/Table List (about 21 figures / 6 tables, core picked)

Figure/ Table	What it conveys	Axes/baseline
Fig 1 (hero)	loss falling across compute/data/ parameters three panels	log-log; color scale = parameter count
Fig 4 left	$L(N,D)$ with data and model	multiple curves, corresponding to Eq (1.5)
Fig 4 right	learning curves unified	after S_{min} normalization models stack into one
Fig 5	shape insensitivity	fix N, vary dff/dmodel, head dim, aspect ratio, loss changes only a few percent
Fig 6	methodology comparison: with/without embeddings	left messy/right converging into one—proving the importance of measurement choice
Fig 7	Transformer vs LSTM	with parameters; LSTM falls behind in the late long-context segment
Fig 9	overfitting and $N^{0.74}/D$ normalization collapse	multiple D curves stack proportionally
Fig 13	$L(C_{min})$ fit	labeled "lump = 1-layer→2-layer transition, removed"
Fig 15	contradiction intersection	$L(C_{min})$ and $L(D(C_{min}))$ intersect = extrapolation failure point
Table 1	per-layer parameters/FLOPs	engineering bookkeeping
Table 2/3	$L(N,D)$, $L(N,S)$ fit parameters	exponents + scaling constants
Table 4/5/6	all power-law summaries, fitted values, optimal allocation	for lookup

Caption self-consistency: every caption carries "what the horizontal axis is, what the color scale is, which equation it corresponds to," nearly readable without the body.

2.8 How to Write Data Validation Methods

- **Data:** WebText2 (96GB, 2.29×10^{10} tokens, 6.6×10^8 test tokens reserved); also testing generalization on Books/CC/Wikipedia/Internet Books.
- **Training:** Adam, fixed 2.5×10^5 steps, batch= 512×1024 ; >1B params use Adafactor; 3000-step linear warmup + cosine to zero; 10% dropout; early-stop by test loss.
- **Sample-size span:** $N=768 \rightarrow 1.5 \times 10^9$, $D=22M \rightarrow 23B$.

- **Random seed noise:** " ~ 0.02 ," and thereby deriving the anti-overfitting threshold $D \gtrsim 5e3 \cdot N^{0.74}$ (Eq 4.4).
- **Reproduction caliber:** $C \approx 6NBS$ (forward $2\times$ + backward $2\times$ + ...), 1 PF-day = $8.64e19$ FLOP defined on the spot.

2.9 Layout Standards

Single-column arXiv; contents printed before the body; sections 1–8 two levels (1.1/1.2...); formulas uniformly numbered (1.1)–(6.8) and appendices (A.x)/(B.x); citations use author abbreviations + year numeric keys ([VSP+17], [RWC+19], [MKAT18]); appendices split A/B/C/D, **Section C the standard location for "limitations."**

III. NSA (DeepSeek 2502.11089, 2025)

3.1 Structural Skeleton and Length Allocation

24 pages, single-column LaTeX (NeurIPS style, section numbering "1. 2. 3."). Sections measured:

Section	Title	Pages	Feature
—	Abstract	p1	problem→gap→contribution→results
1	Introduction	p1–3	64k decoding attention takes 70–80% latency; two challenges; Figure 1 dual axes
2	Rethinking Sparse Attention Methods	p3–5	2.1 The Illusion of Efficient Inference; 2.2 The Myth of Trainable Sparsity; 2.3 Native Sparsity as Imperative
3	Methodology	p5–9	3.1 background (attention + arithmetic intensity); 3.2 overall framework (three-branch gating); 3.3 algorithm (compression/selection/sliding window); 3.4 Kernel design
4	Experiments	p9–13	4.1 pretraining config; 4.2 baselines; 4.3 performance (general/long/CoT)
5	Efficiency Analysis	p13–14	5.1 training speed; 5.2 decoding speed
6	Discussion	p14–15	6.1 failed alternative strategies history; 6.2 attention visualization
7	Related Works	p16	fixed sparse/dynamic pruning/query-aware three classes
8	Conclusion	p16	
—	References / Appendix A	p17–24	Appendix A = AIME sample outputs

Allocation observation: methods + systems (3) and experiments + efficiency (4+5) nearly 1:1; Section 2 "critiquing predecessors" independent, **front-loading the design-space derivation**—this is the structural move most worth copying from NSA.

3.2 Abstract Writing

Problem (long context, attention compute expensive) → existing direction (sparse attention) → **dual gap points** (theoretical speedup $\neq$ actual latency; most only do inference, lack training support) → **two innovations** (① arithmetic-intensity-balanced hardware alignment; ② end-to-end trainable) → results (not inferior to / exceeding Full Attention; 64k decode/forward/backward all accelerated).

3.3 Introduction and the "Critique-Based Positioning" Pattern

- **Section 2 naming-style critique:** "The Illusion of Efficient Inference" and "The Myth of Trainable Sparsity." Under each name two concrete charges:
- Illusion: ① Phase-Restricted Sparsity (H2O sparse only at decode, prefill still full; MInference opposite, always one phase not saving); ② incompatible with MQA/GQA (Quest selects KV independently per head, under GQA taking the union → memory access not saving).
- Myth: ① post-hoc sparsity deviates from the pretraining trajectory (top-20% covers only 70% attention); ② non-trainable components (k-means/SimHash cut gradients); ③ backward propagation inefficient (token-level selection breaks continuous memory access).
- **Translating constraints into design axioms:** hardware alignment wanted → blockwise continuous memory + Tensor Core; end-to-end trainable wanted → selection must be differentiable.

3.4 Division of Methods/Experiments/Efficiency/Discussion

- **Methods = three branches:** compression (block aggregation MLP, global coarse) + selection (reusing the compression branch's attention score as block importance, top-n, differentiable, zero extra overhead) + sliding window (local); each with independent K/V to prevent shortcuts, gate fusion.
- **Efficiency independent section (Section 5):** performance in Section 4, latency in Section 5, avoiding confusion.
- **Discussion = failure history:** 6.1 lists all tried key-clustering, aux-loss selection, heuristic selection, cold-start and draws loss curves (Fig7), proving "why not those."

3.5 Transferable Argumentation Paradigms (this paper distills 9 items)

1. **Hardware-aligned constraints → architecture design:** first explaining arithmetic intensity/roofline (prefill/fwd compute-bound, decode memory-bound), making the algorithm obey each phase's bottleneck—blockwise continuous memory access, GQA intra-group shared KV blocks.
2. **End-to-end trainable as the core selling point:** critiquing k-means/SimHash non-differentiable → NSA directly derives selection importance from the compression branch's intermediate attention score (Eq 8–10), **reusing compute, differentiable, zero extra parameters.**
3. **"Illusion/Myth"-style naming critique:** using the two Section 2 titles to pin down predecessors' methods, forcing out one's own design space.
4. **Three-branch modularization + gating:** global coarse / local fine / neighborhood near, separately ablable; independent K/V preventing gradient crosstalk.
5. **Three-layer baselines + unified budget:** Full Attention (capability ceiling) / Exact-Top (oracle upper bound) / H2O • InfLLM • Quest (SOTA); all sparse methods unified activation token budget (2560) for fairness; **on short sequences the sparse baseline $\approx$ Full, then not compared** (honestly stating where compared, why).
6. **Quality $\times$ efficiency dual-axis reporting:** performance tables (Table1 general / Table2 LongBench / Table3 AIME) + speed figures (Fig6 fwd 9.0 $\times$ /bwd 6.0 $\times$) + theoretical memory access table (Table4 decode 11.6 $\times$, linear in access volume).
7. **Intuition visualization reverse-inferring design:** Fig8 attention heatmap shows blockwise clustering → arguing blockwise selection reasonable; Fig7 three-strategy loss comparison.
8. **Figure 1 one figure dual conclusions:** left "more accurate," right "faster," the hero figure directly endorses "both economical and good."

9. Reproduction config all written (see 3.8).

3.6 Figure/Table List (one by one)

Figure/ Table	What it conveys	Axes/baseline
Fig 1 (hero)	left: General/LongBench/Reasoning Full vs NSA; right: Decode/Forward/Backward acceleration $11.6\times/9.0\times/6.0\times$	grouped bars + speedup ratios
Fig 2	three-branch architecture + three classes of attention masks	green = compute, white = skip; query/activated/evicted/ignored legend
Fig 3	Kernel design	loading Q by GQA group (Grid Loop), inner loop pulling KV blocks, SRAM(green)/HBM(blue)
Fig 4	pretraining loss curves	step 0–60k, Full vs NSA dual lines, NSA lower
Fig 5	needle in a haystack 64k	depth $\times$ position heatmap, all green (100% retrieval)
Fig 6	Fwd/Bwd latency with length	8k–64k bars, labeled $2.1\times\ldots9.0\times/6.3\times$
Fig 7	alternative strategy loss comparison	Full / aux-loss / heuristic / NSA four lines
Fig 8	attention map	blockwise clustering visualization
Table 1	general benchmark 9 metrics + Avg	MMLU...HumanEval; three-line table , NSA bold on most
Table 2	LongBench 11 subsets + Avg	H2O/InfLLM/Quest/Exact-Top/Full/NSA six rows
Table 3	AIME 8k/16k	Full-R 0.046/0.092; NSA-R 0.121/0.146
Table 4	decoding access token counts and expected speedup	8192...65536; Expected Speedup $4\times\ldots11.6\times$

3.7 How to Write Data Validation Methods

- **Base:** GQA+MoE, total 27B/activated 3B, 30 layers, hidden 2560, GQA 4 groups 64 heads, dq=dk=192, dv=128; DeepSeekMoE 72 routed + 2 shared experts top-6, first layer switched to SwiGLU MLP.
- **Sparse hyperparameters:** compression block $l=32$, stride $d=16$, selection block $l'=64$, selected block count $n=16$ (fixed activation 1 initial block + 2 local blocks), sliding window $w=512$.
- **Training:** 270B tokens, 8k length, then YaRN continued training + SFT to 32k; **both sides trained to full convergence for fairness.**
- **Fairness discipline:** sparse baselines unified activation 2560 tokens (StreamLLM caliber: first 128 + 512 local); speed comparison both sides using Triton (NSA kernel vs FlashAttention-2); AIME 16 samples per question, temp 0.7, top-p 0.95 averaged.
- **Hardware:** $8\times$ A100.

3.8 Layout Standards

Single-column LaTeX; section numbering style "1. 2. 3."; **booktabs three-line tables** (top/below-header/bottom three horizontal rules, exactly the style this volume P0–P12 should align to); author-year citations ("DeepSeek-AI, 2025"); formula line numbers (1)–(12); **no independent Limitations section**

(limitations scattered in 6.1 failure history and various "we defer..."); Appendix A holds reviewable complete AIME reasoning samples.

Part Two • Five Rounds of Thinking Drills (mandatory trace)

Each round fixed writing: **method / input / output / concrete impact on later rewriting.**

Drill ① Logical Deduction: Restoring Each Paper's Argument Chain

- **Method:** splitting each paper into the quadruple "premise → mechanism → evidence → conclusion," asking ring by ring "what if the previous ring were missing."
- **Input:** the three PDFs' body + figures/tables.
- **Output:**
 - **AlexNet chain:** small data insufficient (premise) → large data + large model need a trainable structure (CNN priors, mechanism) → but large models overfit/too slow (contradiction) → ReLU speedup, dropout/augmentation anti-overfitting, GPU implementation landing (means) → comparison table proving crushing SOTA (evidence) → depth cannot be removed (sublimation).
 - **Kaplan chain:** loss varies with scale (premise) → fixing variables to measure single-variable power laws (mechanism) → power laws not bending over >6 orders (evidence) → merging into $L(N,D)$ and $L(N,S)$ (synthesis equations) → taking derivatives for optimal allocation (corollary) → self-referential intersection = extrapolation boundary (closing).
 - **NSA chain:** long-context attention expensive (premise) → predecessors' sparse theoretically fast but actually not, and untrainable (critique) → hardware alignment + differentiable selection = three branches (mechanism) → same-budget comparison proving "more accurate and faster" (evidence) → blockwise clustering visualization endorsing (mechanism self-consistency).
- **Impact on later rewriting:** every UDOS paper must write in one sentence the "premise → contradiction → mechanism → evidence" closed loop; **if a paper lacks the "contradiction" ring, it is a P0 structural defect.** Kaplan's "self-referential intersection" demonstrates how to actively bound a conclusion.

Drill ② Reverse Thinking: From the Conclusion Back to What It Must Prove, What Counterexample It Most Fears

- **Method:** assuming the conclusion true, back-inferring "what data would a reviewer use to attack it."
- **Input:** the three headline conclusions.
- **Output:**
 - **AlexNet** conclusion "depth matters" → must prove "removing layers loses points" and "not accidental from data/init." Most feared counterexample: **removing some layer not losing points** (if true, the depth narrative collapses). The authors' defense = Fig3 points out specialization independent of initialization, discussion quantifies "middle-layer removal $\approx -2\%$." → **lesson: whenever claiming "X matters," must give "the cost of removing X."**

- **Kaplan** conclusion "power laws extrapolable" → must prove no bending over many orders. Most feared counterexample: **upper-end bending / collapsing on distribution change**. The authors' defense = Fig8 cross-distribution parallel translation, 6.3 self-disclosed intersection, Section C 6 Caveats. → **lesson: an extrapolation paper's credibility comes from "actively reporting the failure point," not "claiming it holds forever."**
 - **NSA** conclusion "sparse not harming capability and faster" → must prove under the **same sparse budget** not inferior to Full, and the kernel really fast. Most feared counterexample: ① secretly adding more activation tokens; ② only comparing short sequences; ③ speed from an unfair backend. The authors' defense = unified 2560 token budget, both sides using Triton, taking Exact-Top as the oracle upper bound. → **lesson: "faster" must be same hardware same backend same budget; "not harming capability" must have an oracle upper bound as the ceiling.**
 - **Impact on later rewriting:** after each UDOS conclusion sentence, must add a line "**most likely counterexample + how this paper defends**" (scale/saturation papers like P0/P0b/P9 especially need Kaplan-style failure points).
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Drill ③ Divergent Thinking: Paradigm → Paper-by-Paper Mapping Table for This Volume's 15 Papers

- **Method:** mapping the paradigms distilled from the three papers one by one onto UDOS P0c / P0 / P0b / P1–P12.
- **Input:** this volume's contents (P0c metacognition, P0 spatial structure, P0b structure + scale Scaling Laws; P1 layered hybrid topology million-level extrapolation; P2 BFT-lite stopping decisions; P3 circuit breaker RCT; P4 submodular active collection; P5 conformal prediction; P6 identifiability; P7 embodied takeover rate; P8 Gaussian splat Sim2Real; P9 quality saturation law; P10 TransferBundle handoff loss; P11 task market Byzantine slashing; P12 evidence grading reproducibility).
- **Output (paper-by-paper mapping):**

Paper	Paradigm most worth copying	Concrete action
P0c metacognition/ boundary calibration	Kaplan ① conclusion map + ⑧ Caveats	introduction first throwing N "self-cognition boundary" conclusion bullets (with section numbers); dedicated "calibration failure region" subsection, explicitly saying in which cases boundary estimates collapse
P0 spatial structure is intelligence	AlexNet ① engineering as contribution + ⑤ ablation	taking each "spatial constraint" design decision as a contribution, giving removal costs one by one; using Fig6-style "with/without some structure" comparison
P0b structure + scale Scaling Laws	Kaplan full set (controlled variables + power law + exponent table + Caveats)	directly copy: N/D/C three-factor power law, appendix exponent summary table, 6 limitations; this is the paper most isomorphic to Kaplan
P1 layered hybrid topology million-level extrapolation	Kaplan ① controlled-variable extrapolation + ⑤ theory↔empirics reconciliation	fix topology vary scale to fit coordination-complexity power law; first derive the analytic form then compare with measurement
P2 BFT-lite stopping decisions	NSA ③ naming critique + AlexNet ③ fairness self-disclosure	first naming existing stopping rules' "illusion"; actively disclosing under which network partitions this protocol is biased
P3 circuit breaker RCT	AlexNet ② per-trick quantified ablation + ⑥ qualitative figures	each circuit-breaker strategy giving completion-rate increment ($\pm x\%$); with before/after fault-injection comparison visualization
P4 submodular active collection	AlexNet ② ablation + NSA ⑤ three-layer baselines	comparing random/heuristic/submodular three sampling, unified budget; giving sampling-rate-gain curves
P5 conformal prediction fan	Kaplan ⑦ actively finding failure + ② range selling point	the fan unfolding with the uncertainty range; explicitly saying when coverage probability fails under distribution drift
P6 observability/ identifiability	Kaplan ⑨ symbol definition discipline + ⑥ theory reconciliation	first centrally defining hidden/observed variables and Fisher information; analytic identifiability conditions vs numerical experiments
P7 embodied takeover rate	NSA ⑥ quality $\times$ efficiency dual axes + ⑨ reproduction config	single-sided "semantic layer/action prior" two-way ablation; takeover rate table + reproduction params all written
P8 Gaussian splat Sim2Real	NSA ④ modular branches + ⑦ intuition visualization	multiview splat each component ablatable; rendering-real-shot margin heatmap
P9 quality saturation law/homogeneity failure	Kaplan ⑦ self-referential contradiction + ⑧ Caveats	drawing "expansion diminishing-return/saturation intersection," explicitly saying this intersection sensitive to exponent errors; limitations standalone
P10 TransferBundle handoff loss	AlexNet ④ layer-removal cost + NSA ⑦ mechanism visualization	removing the handoff channel to measure information loss; pre/post handoff feature similarity heatmap
P11 task market Byzantine slashing	NSA ③ naming critique + ⑤ same-budget baselines	naming the "no-slashing" market's failure modes; honest node/Byzantine ratio unified comparison
P12 evidence grading reproducibility	Kaplan ④ parameter table + NSA ⑨ reproduction config + AlexNet ⑤ comparison table	making evidence grading a lookup table; system config, seeds, compute all written

- **Impact on later rewriting:** this table is the "paradigm allocation list" during Stage B/C revision—each paper first checks the table, confirming the corresponding paradigm implemented before entering language polishing.

Drill ④ Critical Thinking: Each Paper's Method Limitations, Questionable Points, Our Misuse Risk

- **Method:** standing in the reviewer's position, picking hard flaws one by one.
- **Input:** the three full papers + figures/tables.
- **Output:**
- **AlexNet limitations and misuse risk:** ① the dual-GPU comparison "biased in favor of one-GPU" (authors self-disclosed footnote 2), i.e. the 1.7% improvement may be exaggerated by unfair comparison; ② the ReLU speedup conclusion only demonstrated on one small CIFAR-10 network, extrapolating to ImageNet is "narrative extrapolation"; ③ LRN later proved of little use (replaced by BN)—**historically eliminated tricks should not be copied as "required components"**; ④ misuse risk: if UDOS takes "some small engineering trick bringing $\pm 0.x\%$ " as the core contribution without ablation, it will look thin.
- **Kaplan limitations and misuse risk** (its own Caveats already list, but residuals remain): ① the power law only observed at $N \leq 1.5B$, $D \leq 23B$, extrapolating to GPT-4 level is **aggressive extrapolation** (authors self-admit the intersection $C \sim 1e4$ PF-days floating up to an order of magnitude); ② later Chinchilla proved its "large models more important" biased in data ratio—pure loss scaling $\neq$ downstream-task optimum; ③ the fit sensitive to artificial cleaning such as "removing 1-layer networks, removing the smallest datasets"; ④ misuse risk: if UDOS carries Kaplan's concrete exponents (0.076 etc.) as cross-domain constants, it is a hard flaw; exponents must be self-fitted, only the method borrowed not the values*.
- **NSA limitations and misuse risk:** ① **no independent Limitations section**—27B/260B token single-point validation, not reproduced below 7B or above 100B; ② "NSA exceeds Full Attention" only holds on its own 27B GQA+MoE base, cross-architecture generalization unknown; ③ AIME improvement (+0.075/+0.054) based on 16 samples, small problem count (AIME 24 only ~30 problems), **huge variance yet no error bars given**; ④ the speedup is a Triton implementation, switching to a CUDA backend may not hold; ⑤ misuse risk: if UDOS copies "sparse/compression lossless capability" without same-budget ablation, it falls into the "illusion" NSA Section 2 itself critiques.
- **Impact on later rewriting:** UDOS's 15 papers **each must have an explicit limitations section** (even learning from NSA must be more honest than it), and **whenever citing exponents/speedups/scores, must mark sample size, hardware, variance**, forbidding single-point improvements without error bars written as conclusions.

Drill ⑤ Summary Review: Distilling the Unified UDOS Paper Writing and Layout Standard v2

- **Method:** merging the first four rounds' actionable items into one hard Stage B/C standard.
- **Input:** the three papers + four drill conclusions.
- **Output standard** (this is the Stage B/C execution standard):

A. Structure 1. Abstract fixed five beats: task → headline number → scale anchor → technical keywords → one counterintuitive conclusion. 2. The introduction ending must have an "**N specific contributions**" list, each with a parenthetical source section (Kaplan-style conclusion map). 3. Methods ordered by "authors' self-estimated importance," most important decisions first (AlexNet 3.1–3.4 ordering convention). 4. **Must have an explicit limitations/failure-point section** (learn Kaplan Section C; NSA lacking this section is a negative example).

B. Argumentation 5. Claiming "X matters/effective," must give "the cost of removing X" (AlexNet layer removal) or "same-budget comparison" (NSA unified token budget). 6. Extrapolation conclusions must give "failure point/intersection + magnitude uncertainty" (Kaplan 6.3). 7. Theory derivation and measurement **bidirectionally reconciled**, when consistent write "excellent agreement," when inconsistent write the reason. 8. Actively disclosing biases against oneself (AlexNet footnote 2).

C. Figures/Tables 9. Three required figures: a hero dual-axis figure (capability × efficiency, NSA Fig1), a mechanism visualization figure (attention/features/heatmap), a comparison table (three-line table, italics = others, bold = own best). 10. Caption self-consistency: horizontal axis, color scale, corresponding equation, applicability boundary all written into the caption, readable without the body. 11. Comparison tables use booktabs three-line tables (NSA style), not plain small tables.

D. Data and Reproduction 12. Reproduction config all written: base scale/layers/dimensions/hyperparameters/data volume/hardware/optimizer/learning rate/seeds/random noise magnitude. 13. Every improvement number marked with sample size, hardware, variance or error bars; single-point small-sample results not written as settled conclusions. 14. Cross-dataset/cross-distribution generalization separately tested (Kaplan Fig8-style parallel translation).

E. Layout 15. Single-column LaTeX; two-level section numbering; formulas uniformly numbered; references numbered or author-year, one choice unified across the volume; appendices centrally hold "summary tables + supplementary experiments + limitations."

- **Impact on later rewriting:** this standard's 15 items = Stage B revision checklist, Stage C final acceptance standard; any paper not satisfying items 4/6/9/13 is sent back.

Part Three • Delivery Notes

- All content of this report comes from PyMuPDF full-text extraction + page-rendered image reading of the three PDFs, not from memory.
- Transferable paradigm counts distilled from the three: **AlexNet 9 / Kaplan 10 / NSA 9**.
- Each of the five drills' one-sentence conclusion: ① logical deduction—every UDOS paper must have the "premise → contradiction → mechanism → evidence" closed loop, lacking the contradiction ring = P0; ② reverse thinking—whenever claiming "X matters/extrapolable," must self-report "most likely counterexample and how this paper defends"; ③ divergent thinking—produced the 15-paper paper-by-paper paradigm mapping table (P0c/P0/P0b/P1–P12); ④ critical thinking—exponents/speedups/scores must carry sample size and variance, every UDOS paper must have a limitations section more honest than NSA; ⑤ summary review—solidified into the 15-item UDOS Writing and Layout Standard v2 as the Stage B/C acceptance standard.
- Synchronous action: the three bibliographic records were appended to Master References Library.md under "source = user attachment PDF original, verification date 2026-09-20" (see the new block at that file's end, append only, other content unchanged).